The RX Path

How the node's receive audio gets from Asterisk into Node Talker, through the frame queue, the cleanup stage, and out to each listener, and what happens when the queue runs behind.

FEED HOOK · FRAMES · CLEANUP · LISTENERS

SUMMARY · READ-ONLY REVIEW

From Asterisk to a UDP port

1.1 The hook in rpt.conf

Enabling the receive feed for a node writes two lines into that node's block in rpt.conf: a note saying the line sends the node's audio to Node Talker, and an `outstreamcmd` line. The command runs the tool's own feed mode with Python, and passes the UDP port. The code refuses to add the hook if the node already has an `outstreamcmd` that is not Node Talker's (for example a Broadcastify feed), since a node takes only one.

1.2 The feed process

The feed process reads standard input in blocks of up to 4096 bytes and sends each block as one UDP datagram to 127.0.0.1 on the chosen port. It does no processing. It stops when the input ends or the output fails. A restart of Asterisk is needed after the hook is added or changed.

Frames and the queue

SETTING	VALUE	MEANING
Frame	160 samples, 320 bytes, 20 ms	One unit of audio, 16-bit PCM at 8 kHz
Input queue limit	25 frames (0.5 s)	Above this, the oldest frames are dropped and counted as over-limit drops
Input trim point	10 frames (0.2 s)	When the pump falls behind, one frame a tick is dropped and counted as a trim drop
Listener queue	150 frames (3 s)	Each listener's queue; a full queue is counted as an overrun
Active level	-50 dBFS	A frame louder than this counts as activity
Activity hold	0.6 s	How long the channel stays "active" after the last loud frame

2.1 The pump

A paced loop runs a tick every 20 ms. Each tick takes one frame off the queue, measures its level, and checks for transmit activity. It then runs the cleanup stage (Part 3) if any listener is connected or a tone test is recording. Without listeners, the tick only keeps the level and burst counts up to date, so an idle node costs little.

2.2 Bursts and kerchunks

A burst is a run of loud frames. A burst at least 0.1 s long is logged as a possible kerchunk when it is shorter than a learned threshold (a short key-up, often a keying tail). A burst that starts while Node Talker is keying, or within one second after it unkeys, is marked as the node's own transmission, and is not counted as a kerchunk.

2.3 Delay for kerchunks

The setting `kerchunk_delay_ms` holds back all audio by a fixed time. It is 0 (off) or 100 to 1000 ms. When it is on, bursts that were marked as short kerchunks are dropped before they reach listeners, so the listener does not hear them.

Filter, gain, AGC and gate

The cleanup stage works on each frame in this order. Every stage can be switched off, and when all are off the stage only applies the RX gain.

3.1 High-pass filter

When `rx_hpf` is on, two cascaded biquad sections form a fourth-order high-pass at 300 Hz. It removes low-frequency rumble. Its filter memory is kept between frames, so there is no click at each frame edge.

3.2 Notch filter

When `rx_notch_hz` is set (60 to 300 Hz), a narrow notch removes one tone, usually a CTCSS tone that rides on the audio. The notch has a Q of 25 at an 8 kHz sample rate.

3.3 Automatic gain (AGC)

When `rx_agc` is on, the gain follows the speech level toward a target of -20 dBFS. The gain can range from -6 dB to +18 dB. It rises slowly (0.06 dB a frame) and falls faster (up to 1.5 dB a frame). After a long idle period, it moves quickly for 25 frames to catch up with a new talker.

The fixed RX gain (`rx_gain_db`, -12 to +12 dB) is added on top of the AGC gain, when the level settings are active (Part 4).

3.4 Gate

The gate (`rx_gate`) decides whether a frame is passed or silenced. It has three settings:

SETTING	MEANING	CLOSE BELOW	HOLD
0	Off, every frame passes	n/a	n/a
1 (default)	Gentle: closes only on a quiet channel, or when noise is flat and steady	-55 dBFS	20 frames
2	Strict: closes sooner and treats noise more firmly	-48 dBFS	12 frames

The gate looks four frames ahead (80 ms), so it can open before the speech starts and keep the first syllables. When it closes, the audio is faded out, and when it opens, it is faded in. A frame of noise that is flat and steady counts toward closing the gate, even above the close level.

3.5 Quiet floor

When a frame is quieter than -65 dBFS, the filters are reset to silence, and the gate is driven as though the channel were empty. After 150 idle frames (3 s), any held audio is cleared.

3.6 Limiting

A gain that would push a sample above 29000 is passed through a soft curve, so loud peaks round off rather than clip hard.

Each listener gets its own queue

4.1 Delivery

After the cleanup stage, each frame is sent to every listener's queue. A listener that asked for 16-bit PCM (`fmt=s16`) gets the frame as it is. Any other listener gets it as u-law, which halves the data. Silence is sent as a pre-built u-law silence frame.

4.2 Joining

A listener connects over HTTP and is given a slot. Up to `max_listeners` (default 3) listeners may connect. A request beyond that is refused with 503 and a message that every slot is in use. A client name is recorded from the request's user agent, so the health reports can say what kind of device is listening.

4.3 Slow listeners

Each listener's socket send buffer is 64 KB. A listener that falls behind has its queue filled, and frames are dropped for that listener only. The other listeners are not held back.

The health readings

Node Talker keeps a health record of the receive path, for the Data card download and for the operator. It notes:

- **Gaps and jitter** between packets from the feed, and how late the pump runs.
- **Write times**, to see if a listener's socket is slow.
- **Drops**, split into over-limit, trim and listener overruns.
- **Network and Wi-Fi** readings for each listener's link, where they are available.

A verdict combines these into one reading for the operator. The thresholds are set in the code (for example 60 ms for a jitter flag), and the numbers appear in the page's Data card download.

What the RX path shows

6.1 Two places drop audio, on purpose

The queues drop audio when the Pi falls behind, so the listener hears a short gap rather than a growing delay. The book should tell operators that a dropout on a listener is often the Pi, not the node.

6.2 The cleanup changes what the node sends

The high-pass, the AGC and the gate all change the sound before the listener hears it. A test that compares the listener's audio with the node's raw audio will show these changes. The book should say which settings were on during a test.

6.3 Not checked

- The exact health thresholds and the verdict rules, which were read only in part.
- The phone-side listener, which uses the same stream and was not traced separately.
- Real audio quality under the gate and AGC settings on a node.